# Week 11

Spring 2027 · Section 017 · MW 7:30–8:50 PM · RRC Room 8210

**Course Level Objectives:**

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Describe and compute confidence intervals. (CLO 6)
- Perform hypothesis testing using statistical methods. (CLO 8)

| Module Objectives | Instructional Materials | Learning Activities | Assessments | Tools |
|---|---|---|---|---|
| **1.** Describe how to generate randomization samples for a given study design. (CLO 8)<br>**2.** Identify the centers of the bootstrap and randomization distributions for given data. (CLO 5)<br>**3.** Use a 95% confidence interval to reach a two-tailed test conclusion at α = 0.05. (CLO 6, CLO 8)<br>**4.** Determine whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8) | e-Text: 4.5 Making Connections<br><br>Interactive Lecture: 4.5 Making Connections<br><br>Slides: 4.5 Making Connections | Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples<br><br>Your Turn: CI or Test? (smart-speaker ownership) | Online Homework: 4.5 Making Connections (due Sun Apr 11, 11:59 PM)<br><br>Take-Home Quiz 7 (due Mon Apr 12, 11:59 PM) – covers 4.4, 4.5, 5.1 (all objectives)<br><br>Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives)<br><br>Test 3 Study Guide (all objectives)<br><br>Practice Test 3 (optional; due Thu Apr 29) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Respondus |
| **5.** Explain when a normal distribution can replace a randomization distribution. (CLO 5)<br>**6.** Compute a standardized z statistic. (CLO 8)<br>**7.** Find a p-value as a tail area under the standard normal curve. (CLO 3, CLO 8) | e-Text: 5.1 Hypothesis Tests Using Normal Distributions<br><br>Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions<br><br>Slides: 5.1 Hypothesis Tests Using Normal Distributions<br><br>StatKey – Theoretical Distributions (Normal) | Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value<br><br>Your Turn: z-Tests (right-, left-, and two-tailed) | Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Apr 11, 11:59 PM)<br><br>Take-Home Quiz 7 (due Mon Apr 12, 11:59 PM) – covers 4.4, 4.5, 5.1 (all objectives)<br><br>Test 3 (Fri Apr 30 – Sun May 2, proctored) – covers 4.4–6.5 (all objectives)<br><br>Test 3 Study Guide (all objectives)<br><br>Practice Test 3 (optional; due Thu Apr 29) (all objectives) | WileyPLUS<br>StatKey<br>Blackboard<br>Respondus |

**Also this week:** Test 2 Retake – Mon Apr 5 – Sun Apr 11 (optional; higher score counts)